Name: ______________________________ **/ 25**

30 minutes maximum. No aids (book, calculator, etc.) are permitted. Show all work and use proper notation for full credit. Answers should be in reasonably-simplified form. 25 points possible.

1. Consider the curves $y = x^2$, $y = -x + 2$, and $y = 0$.

   (a) (4 points) Sketch the region bounded by these curves. Be sure to clearly label the curves and any important points. Show any relevant work.

   (b) (6 points) Find the area of the region bounded by the curves.

2. Consider the region bounded by the curves $y = x^3$, $y = 8$, and $x = 0$.

   (a) (4 points) Sketch the region. Be sure to clearly label the curves and any important points. Show any relevant work.

   (b) (6 points) Find the volume when the region is rotated around the y-axis.

   (c) (5 points) Set up, but **do not evaluate** an integral that computes the volume when the region is rotated around the x-axis.